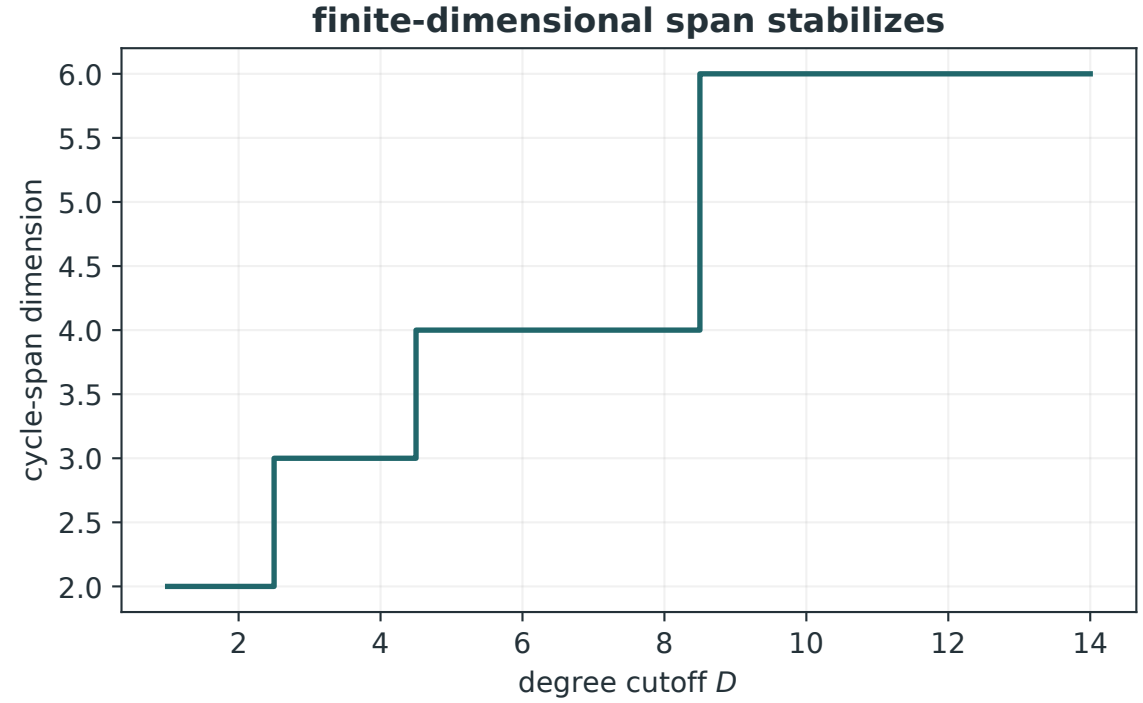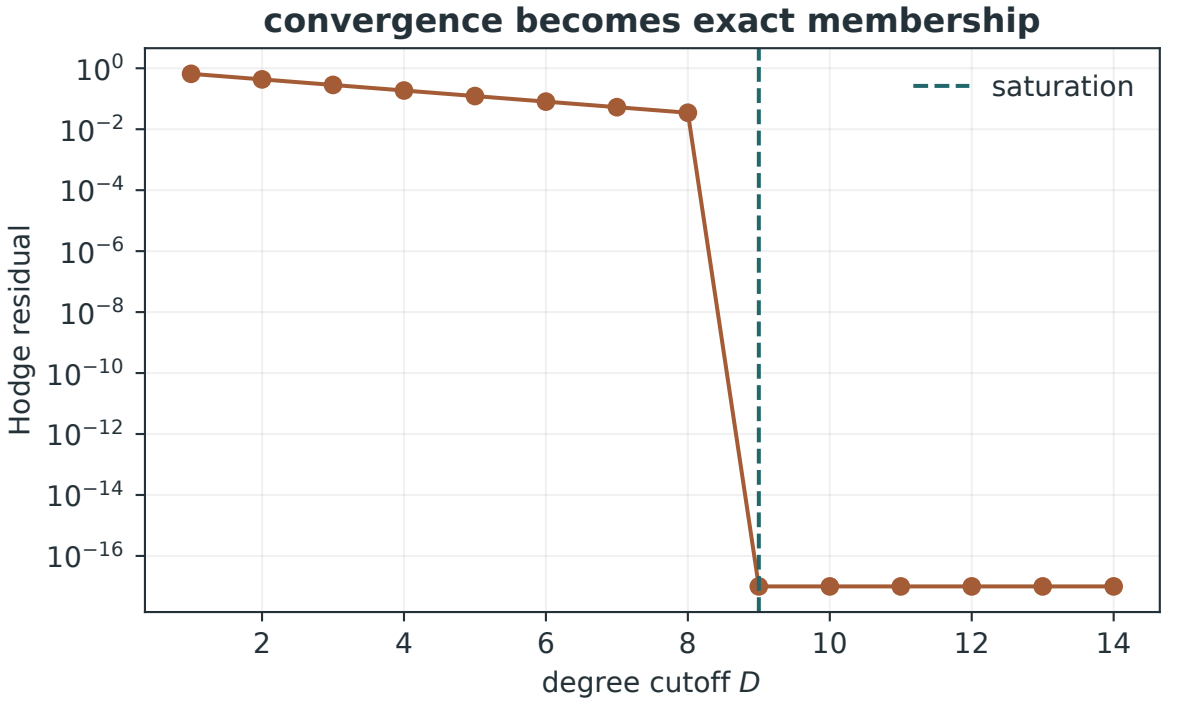Nested bounded-degree cycle spaces can change rank only finitely many times